

Retail Customer Analysis

Phase 1 — Data Exploration & Quality Assessment

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Overview

This first phase examines a real-world sales dataset from a UK-based online retailer covering two years of trading (December 2009 to December 2011). The goal was not to draw business conclusions yet, but to understand what the data contains, how much of it is usable, and what needs to be corrected before any analysis. Think of it as a full inventory check before opening the shop — making sure the records are complete, accurate, and trustworthy.

The Dataset at a Glance

1.07M TRANSACTION RECORDS	53,628 UNIQUE ORDERS	5,942 CUSTOMERS	5,305 PRODUCTS	43 COUNTRIES
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Trading period: 1 December 2009 – 9 December 2011.

Key Findings

1. Nearly a quarter of records cannot be linked to a customer.

243,007 records (22.77%) have no customer identifier. These are likely guest checkouts or data-entry gaps. They remain valuable for product and revenue analysis but cannot be used for customer-level work such as loyalty, lifetime value, or retention — which will therefore be based on the 824,364 identifiable records.

2. The data is broadly healthy, with only minor correctable issues.

Returns and cancellations account for about 2% of records, invalid prices for under 1%, and missing product names for under half a percent — all normal for retail data and straightforward to address.

3. A small number of duplicate records were detected.

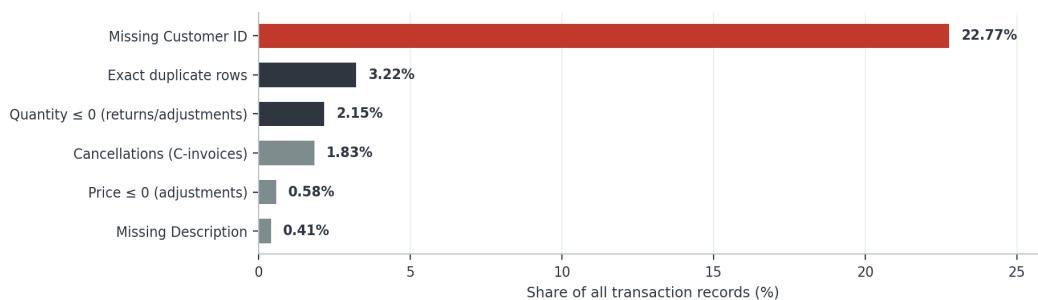
34,335 records (3.22%) are exact duplicates and will be removed so that sales are not counted twice.

4. The dataset is heavily concentrated in the United Kingdom.

The UK accounts for the large majority of activity across all 43 countries, which is expected for a UK retailer and will inform how international results are interpreted.

Data Quality Snapshot

Each issue shown as a share of all records. Categories overlap (a record can be both a cancellation and a negative quantity), so they are not additive.



What's Next — Phase 2

Phase 2 cleans the data based on this assessment: removing duplicates, separating genuine sales from returns and adjustments, and correcting invalid entries. The cleaned data will be organised into two views — all transactions (for product and revenue analysis) and identifiable customers (for loyalty and lifetime-value analysis) — then loaded into a database for deeper analysis.